ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 22-Sep-2026

Site: Test Customer Tech | Location: Bengaluru, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
5.04 kW	**8,720 kWh**	**Rs. 237,450**	**4Y 5M**
5.50 kWp Installed	CUF 18.1% | PR 75%	Rs. 2.37 Lakhs	Profit: Rs. 25.83L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Test Customer Tech
Location / Address	Bengaluru, Karnataka
Contact Email Address	test@example.com
Design Path Selected	Rooftop Area Analysis
System Target Capacity	5.04 kW (Single LT Installation)
Total Installed Array Rating	5.50 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	10 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	600.00 Sq.Feet (55.74 sq.m)
Utilized Usable Rooftop Area	504 Sq.Feet (46.82 sq.m)
First Year Annual Generation	8,720.48 kWh
25-Year Cumulative Generation	193,750.45 kWh
Estimated Payback Period	4 Years & 5 Months
Post-Payback Profitability (25 Yrs)	Rs. 2,583,465.43 (INR 25.83 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 1,146,337.15 (INR 11.46 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 158 Tonnes CO2 | Tree Equivalence: 252 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Main Block (Rooftop Array)
Gross Rooftop Area	600.00 Sq.Feet (55.74 sq.m)
Utilized Effective Rooftop Area	504 Sq.Feet (46.82 sq.m)
Building Plant Target Capacity	5.04 kW
Solar PV Module Rating	550 W N-Type TOPCon
Number of Solar Modules	10 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	4 kW, 230V 1-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (4 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	4 kW Grid-Tied Inverter (Single-Phase / Three-Phase)
Certified Top-Tier OEM Models	Waaree W3.1/3-PH, Luminous NXi 340
Derated AC Design Power (70% Factor)	3.85 kW AC (from 5.04 kWp DC)
AC Output Voltage & Grid Phase	230V, 1-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx6 Sq.mm
DC Array Isolator Switch	25A, MCB Type, 2P 230V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	16.74 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx6 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	32A, 230V, 2PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W N-Type TOPCon	10 Nos.	Rs. 100,000
02	On-grid inverter	4 kW, single-phase (Waaree W3.1/3-PH or equiv.)	1 No.	Rs. 44,000
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 25,000
04	DC solar cable	4-6 sq.mm copper, PV-rated	60 m	Rs. 4,500
05	AC cable	6 sq.mm copper	35 m	Rs. 3,850
06	MC4 connectors	PV-rated connector pairs	10 pairs	Rs. 3,000
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 7,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 7,000
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 3,500
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 7,100
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 25,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 237,450.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 500 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 1,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 4,500 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 3,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 237,450 | ESCOM Utility Budget: Rs. 9,012 | Grand Total Outlay: Rs. 246,462 (Rs. 2.46 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 237,450	**4Y 5M**	**Rs. 25.83L**	**Rs. 11.46L**
12-Item BOM Basis	Break-Even Year	Rs. 2,583,465	Alternative Asset

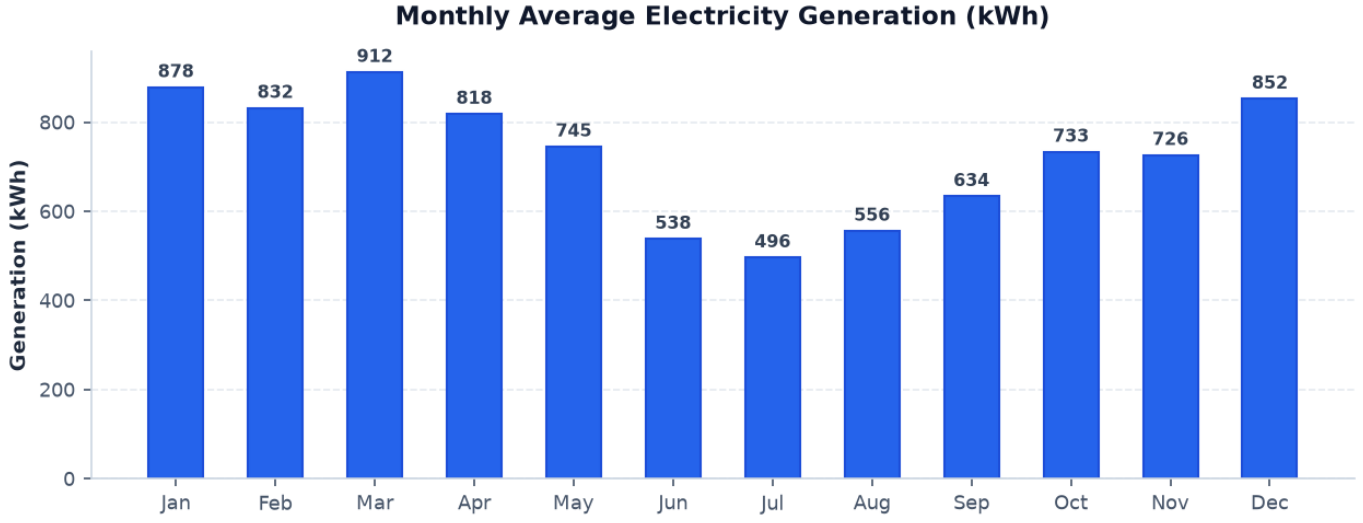
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	8,720	Rs. 8.05	70,200	3,024	67,176	67,176
02	8,633	Rs. 8.45	72,951	3,024	69,927	137,103
03	8,547	Rs. 8.87	75,811	3,024	72,787	209,890
04 *	**8,461**	**Rs. 9.32**	**78,861**	**3,024**	**75,837**	**285,727**
05	8,377	Rs. 9.78	81,926	3,024	78,902	364,629
06	8,293	Rs. 10.27	85,170	3,024	82,146	446,775
07	8,210	Rs. 10.78	88,506	3,024	85,482	532,257
08	8,128	Rs. 11.32	92,010	3,024	88,986	621,242
09	8,047	Rs. 11.89	95,676	3,024	92,652	713,894
10	7,966	Rs. 12.48	99,420	3,024	96,396	810,290
11	7,887	Rs. 13.11	103,394	3,024	100,370	910,660
12	7,808	Rs. 13.76	107,435	3,024	104,411	1,015,071
13	7,730	Rs. 14.45	111,694	3,024	108,670	1,123,741
14	7,652	Rs. 15.18	116,164	3,024	113,140	1,236,881
15	7,576	Rs. 15.93	120,684	3,024	117,660	1,354,540
16	7,500	Rs. 16.74	125,552	3,024	122,528	1,477,068
17	7,425	Rs. 17.57	130,459	3,024	127,435	1,604,504
18	7,351	Rs. 18.45	135,624	3,024	132,600	1,737,103
19	7,277	Rs. 19.37	140,962	3,024	137,938	1,875,042
20	7,205	Rs. 20.34	146,541	3,024	143,517	2,018,559
21	7,133	Rs. 21.36	152,351	3,024	149,327	2,167,886
22	7,061	Rs. 22.43	158,383	3,024	155,359	2,323,245
23	6,991	Rs. 23.55	164,629	3,024	161,605	2,484,850
24	6,921	Rs. 24.73	171,149	3,024	168,125	2,652,975
25	6,851	Rs. 25.96	177,865	3,024	174,841	2,827,815

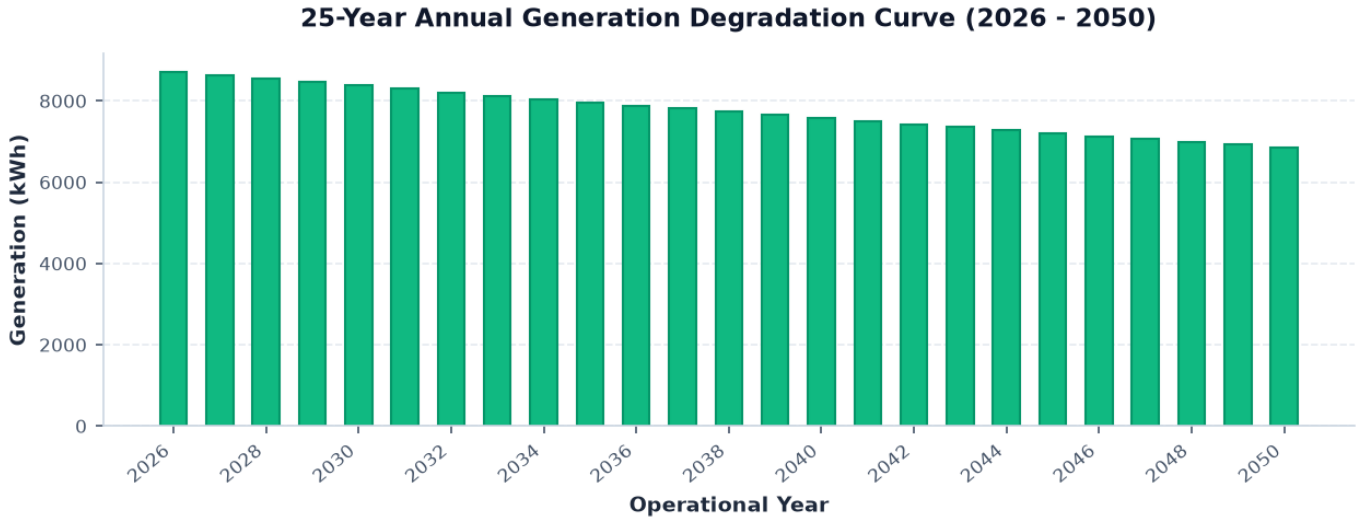
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1730 kWh/kWp	75.0%	18.1%	1.0% / year
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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ON-GRID SOLAR PV SYSTEM

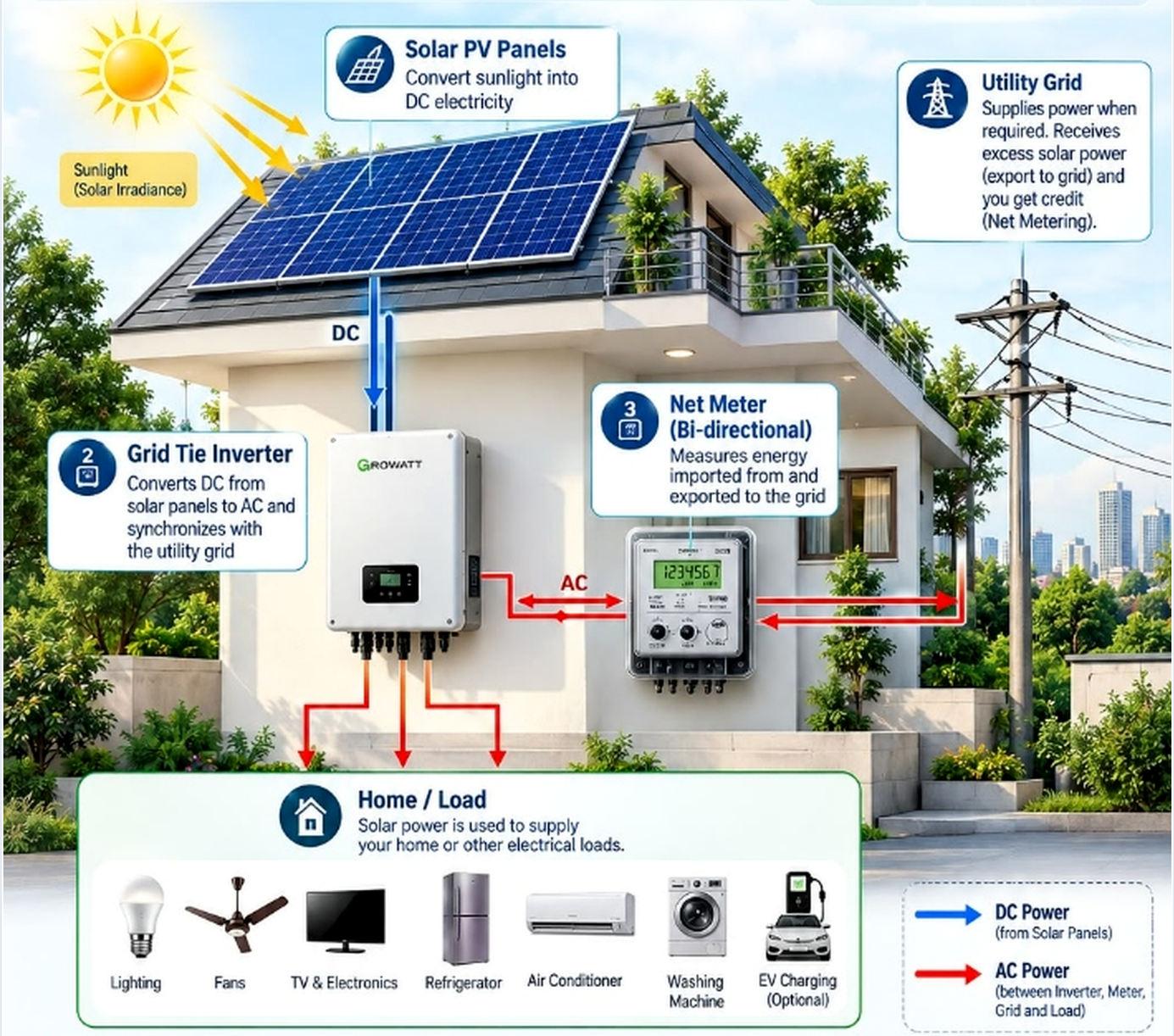
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow